

# Dinusanth Surendran

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## EXPERIENCE

### Software Developer

RBC Capital Markets

Jan 2026 – Present  
Toronto, ON

- Engineering within **Quantitative Technology Services** (Canadian Central Risk Book & Electronic Market Making team) using Spring Boot and Java; supporting quantitative models and financial data infrastructure for capital markets
- Authored deployment automation scripts and consolidated parallel project instances, cutting total execution time from **1 instance/min (unbounded)** to **<2.5 min total** by parallelizing **all 7** instances into a single optimized run
- Built an **end-to-end automated** deployment pipeline, automating SSH-based deployment of scripts, configs, and Spring Boot services across repositories; reduced manual deployment time by 30 minutes per release.

### Software Engineer Intern

LB Connect

Jul 2025 – Feb 2026  
Remote

- Promoted to codebase gatekeeper** within 2 months; approved all backend PRs and managed production releases, maintaining a **<5% bug regression rate** post-release
- Partnered directly with CTO on sprint planning and system architecture, driving decisions across a fast-moving production codebase in PHP

### Automation Analyst Co-op

RBC

May – Dec 2025  
Toronto, ON · Hybrid

- Built and deployed a backend pipeline converting **1,000+ messy financial records/week** into structured CSVs, enabling **3× faster audit reviews** by eliminating parsing failures
- Rewrote legacy Python scripts into production-grade systems; reduced manual QA effort by **60%** via CI/CD integration and enterprise-grade automated testing
- Optimized Python, SQL, and Bash scripts for performance across enterprise financial workflows; contributed to Agile bi-weekly deployment cycles

### Founder & Executive Director

Youth Volunteer Engagement Network (Non-Profit)

Feb 2023 – Present  
Remote

- Founded organization from the ground up, drafting **100+ founding and governance documents** and scaling to **28 active members** through structured outreach and partnership programs
- Designing and building a full-stack website and CMS to manage member onboarding, event registration, and internal process workflows end-to-end

## PROJECTS

### Quantitative Backtesting Research Framework

Python, NumPy, pandas, Plotly

Jul 2025 – Present

- Refactored ETF/equity backtesting engine into a **modular, production-grade research framework** with plug-and-play strategy classes, config-driven runs, and structured experiment tracking
- Implemented full risk analytics suite; **Sharpe, Sortino, Calmar ratios**, max drawdown, profit factor, win rate — validated on 5Y multi-factor ETF strategy (CAGR 5.56%, 153 trades, NAV \$130,955)
- Engineered realistic **slippage/commission modeling** and vectorized NumPy computations for high-performance backtesting across large historical datasets

### Enterprise Options Trading Platform

Python, FastAPI, Streamlit, Plotly, Black-Scholes, Monte Carlo

2025

- Built a full-stack derivatives pricing platform using **Black-Scholes, Binomial Tree, and Monte Carlo** simulation (with variance reduction) for European and American options
- Integrated real-time market data to compute and visualize **Greeks (Delta, Gamma, Vega, Theta, Rho)** and interactive 3D pricing surfaces; FastAPI backend with Streamlit frontend

### Fifa World Cup - Predictor

Python, NumPy, pandas

- Python, NumPy, pandas*, Built a Python World Cup 2026 match prediction system combining **Elo ratings, Dixon–Coles Poisson goal modeling, and 12+ calibrated signals** (form, H2H, squad value, injuries, coach, weather, altitude, travel/jet lag, market odds) into a CLI that forecasts 1X2 outcomes, xG, and secondary stats (shots, corners, cards) for all 48 nations.
- Integrated free and optional APIs (Transfermarkt, Open-Meteo, ESPN, The Odds API, international results) with **caching, walk-forward backtesting, and in-sample calibration** on finished matches to keep predictions realistic and auditable via per-factor breakdowns.

## EDUCATION

### McMaster University

B.Tech, Automation Systems Engineering Technology

Hamilton, ON  
Sep 2023 – Apr 2028

**Relevant Coursework:** Instrumentation & Control, Embedded Systems, PLC Programming, CAD for Design

## TECHNICAL SKILLS

**Languages:** Python, Java, C++, SQL, Bash, PHP, JavaScript

**Quant / Finance:** Black-Scholes, Monte Carlo, Binomial Tree, Greeks, Sharpe/Sortino/Calmar, vectorized NumPy/pandas

**Backend & DevOps:** Spring Boot, FastAPI, REST APIs, PostgreSQL, CI/CD, Docker, GitHub Actions, Linux

**Certifications:** Meta Python Programming (2025) · Automate the Boring Stuff with Python – Udemy (2025)